

Limits

Calculus is the study of *differentiation* and *integration* (this is indicated by the Chinese translation of “calculus”). Both concepts of differentiation and integration are based on the idea of *limit*. In this chapter, we use an intuitive approach to consider limits, omitting the more difficult ϵ - δ definition.

3.1 Introduction

In this section, we introduce the idea of limit by considering two problems. The first problem is to “find” the velocity of an object at a particular instant. The idea is related to *differentiation*. The second problem is to “find” the area under the graph of a curve (and above the x -axis). The idea is related to *integration*.

Problem 1 Suppose an object moves along the x -axis and its *displacement* (in meters) s at time t (in seconds) is given by

$$s(t) = t^2, \quad t \geq 0.$$

We want to consider its *velocity* at a certain time instant, say at $t = 2$.

Idea Velocity (or speed) is defined by

$$\text{velocity} = \frac{\text{distance traveled}}{\text{time elapsed}} \quad (3.1.1)$$

(3.1.1) can only be applied to find average velocities over time intervals.

We (still) don’t have a definition for *velocity* at $t = 2$.

To define the velocity at $t = 2$, we consider short time intervals about $t = 2$, say from $t = 2$ to $t = 2 + \frac{1}{2^n}$. Using (3.1.1), we can compute the average velocity over the time intervals $[2, 2.5]$, $[2, 2.25]$ etc.

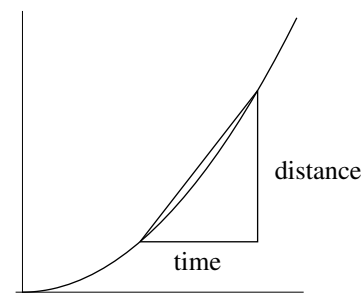


Figure 3.1

n	Time interval	Velocity
1	$[2, 2.5]$	4.5 m/s
2	$[2, 2.25]$	4.25 m/s
3	$[2, 2.125]$	4.125 m/s
4	$[2, 2.0625]$	4.0625 m/s
$\vdots$		

In general, the velocity v_n over the time interval $[2, 2 + \frac{1}{2^n}]$ is

$$\begin{aligned} v_n &= \frac{(2 + \frac{1}{2^n})^2 - 2^2}{\frac{1}{2^n}} \\ &= \frac{4 + 2 \cdot 2 \cdot \frac{1}{2^n} + (\frac{1}{2^n})^2 - 4}{\frac{1}{2^n}} = 4 + \frac{1}{2^n}. \end{aligned}$$

It is clear that if n is very large (that is, if the time interval is very short), v_n is very close to 4. The velocity, called the *instantaneous velocity*, at $t = 2$ is (defined to be) 4.

Problem 2 Find the area of the region that lies under the curve $y = x^2$ and above the x -axis for x between 0 and 1.

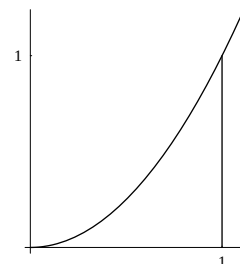


Figure 3.2

Idea Similar to the idea in Problem 1, we use approximation to find/define area. First we divide the interval $[0, 1]$ into finitely many subintervals of equal lengths:

$$\left[0, \frac{1}{n}\right], \left[\frac{1}{n}, \frac{2}{n}\right], \left[\frac{2}{n}, \frac{3}{n}\right], \dots, \left[\frac{n-1}{n}, 1\right].$$

For each subinterval $\left[\frac{i-1}{n}, \frac{i}{n}\right]$, we consider the rectangular region with base on the subinterval and height $\left(\frac{i-1}{n}\right)^2$ (the largest region that lies under the curve). If we add the area of these rectangular regions, the sum is smaller than that of the required region. However, if n is very large, the error is very small and we get a good approximation for the required area.

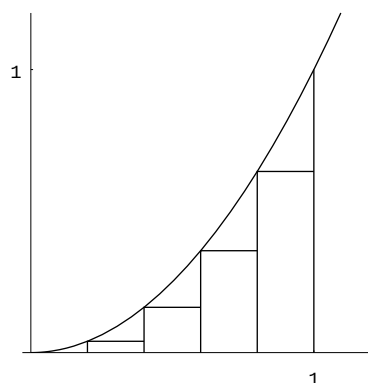


Figure 3.3

The following table gives the sum S_n of the areas of the rectangular regions (correct to 3 decimal places) for several values of n .

In general, if there are n subintervals, the sum S_n is

$$\begin{aligned} S_n &= \frac{1}{n} \cdot 0^2 + \frac{1}{n} \cdot \left(\frac{1}{n}\right)^2 + \frac{1}{n} \cdot \left(\frac{2}{n}\right)^2 + \dots + \frac{1}{n} \cdot \left(\frac{n-1}{n}\right)^2 \\ &= \frac{1^2 + 2^2 + \dots + (n-1)^2}{n^3} \\ &= \frac{n(n-1)(2n-1)}{6n^3} && \text{By Sum of Squares Formula} \\ &= \frac{2n^3 - 3n^2 + n}{6n^3} \\ &= \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2} \end{aligned}$$

n	Sum of areas
2	0.125
3	0.185
4	0.219
$\vdots$	
10	0.285
$\vdots$	
100	0.328
$\vdots$	
500	0.332

It is clear that if n is very large (so that the error is small), S_n is very close to $\frac{1}{3}$.

3.2. Limits of Sequences

Conclusion The area of the region is $\frac{1}{3}$.

Sum of Squares Formula $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Exercise 3.1

- *1. In the first problem: If we consider short time intervals other than $[2, 2 + \frac{1}{2n}]$, for example consider $[2, 2 + \frac{1}{n}]$ or $[2 - \frac{1}{n^2}, 2]$, do we get the same result?
- *2. In the second problem:
 - (a) For each subinterval, we use the value of y at the left endpoint as the height of the rectangular region. If we take the right endpoint instead (we will get a surplus in this case), do we get the same result? How about taking an arbitrary point in each subinterval?
 - (b) In finding the approximations, we divide $[0, 1]$ into equal subintervals. How about dividing it into unequal subintervals?

3.2 Limits of Sequences

In the last section, we obtained formulas for v_n and S_n in Problems 1 and 2 respectively. Each of these formulas gives a *sequence* (which is a special type of function). To consider the behavior of a sequence for large n , we introduce the concept of *limit of a sequence*.

Definition

- A *sequence* is a function whose domain is $\mathbb{Z}_+$ (the set of all positive integers).
- A *sequence of real numbers* is a sequence whose codomain is $\mathbb{R}$.

A sequence of real numbers is a function from $\mathbb{Z}_+$ to $\mathbb{R}$. In this course, we will not consider sequences with codomains different from $\mathbb{R}$. Thus, in what follows, a sequence means a sequence of real numbers.

Let $f : \mathbb{Z}_+ \rightarrow \mathbb{R}$ be a sequence. For each positive integer n , the value $f(n)$ is called the *n th term* of the sequence and is usually denoted by a small letter together with n in the subscript, for example a_n . The sequence is also denoted by $(a_n)_{n=1}^{\infty}$ because if we know all the a_n 's, then we know the sequence.

Sometimes, we represent a sequence $(a_n)_{n=1}^{\infty}$ by listing a few terms in the sequence:

$$a_1, a_2, a_3, a_4, a_5, \dots$$

In the following example, the sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are the sequences obtained in Problem 1 and Problem 2 in the last section respectively.

Example

- (1) Let $a_n = 4 + \frac{1}{2^n}$. The sequence $(a_n)_{n=1}^{\infty}$ can be represented by

$$\frac{9}{2}, \frac{17}{4}, \frac{33}{8}, \frac{65}{16}, \dots \quad (3.2.1)$$

(2) Let $b_n = \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2}$. The sequence $(b_n)_{n=1}^{\infty}$ can be represented by

$$0, \frac{1}{8}, \frac{5}{27}, \frac{7}{32}, \frac{6}{25}, \dots \quad (3.2.2)$$

Remark

- It is not a good way to describe a sequence by listing a few terms in the sequence. For example, in (3.2.1) or (3.2.2), it may not be easy to find a formula for the n th term. Moreover, different people may obtain different formulas. It is better to describe a sequence by writing down a formula for the n th term explicitly.
- To denote a sequence, some authors use the notation $\{a_n\}_{n=1}^{\infty}$ instead of $(a_n)_{n=1}^{\infty}$.

Definition A sequence $(a_n)_{n=1}^{\infty}$ is said to be *convergent* if there exists a real number L such that

(*) a_n is arbitrarily close to L if n is sufficiently large.

Remark Condition (*) means that we can make $|a_n - L|$ as small as we want by taking n large enough. For the sequence $(a_n)_{n=1}^{\infty}$ where $a_n = \frac{1}{2^n}$, we can make a_n arbitrarily close to 0 by taking n large enough. For example, if we want $|\frac{1}{2^n} - 0| < 0.01$, we can take $n > 7$; if we want $|\frac{1}{2^n} - 0| < 0.001$, we can take $n > 10$ etc.

Intuitively, Condition (*) means that if we let n increase without bound (or let n approach “ ∞ ”, an imaginary point very far on the right), the value a_n approaches L . Geometrically, this means that the point (n, a_n) approaches the horizontal line $y = L$ as n increases without bound.

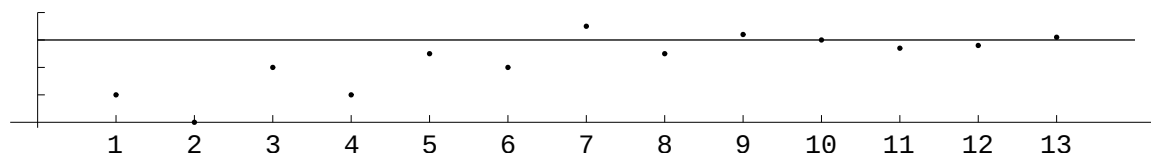


Figure 3.4

Remark For simplicity, instead of saying Condition (*), we will say

(**) a_n is close to L if n is large.

In the definition of “*convergent*”, it is clear that if L exists, then it is unique. We say that L is the *limit* of $(a_n)_{n=1}^{\infty}$ and we write $\lim_{n \rightarrow \infty} a_n = L$.

FAQ Can we just write $\lim a_n = L$, omitting $n \rightarrow \infty$?

Answer For sequences, this will not cause ambiguity. However, for functions, we will consider (in later sections) *limits at infinity* as well as *limits at a point a* (where $a \in \mathbb{R}$). The notations $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow a} f(x)$ have different meanings. $\square$

The following rules can be proved by definition using an alternative method, called ϵ - N method, to describe condition (*). However, the ϵ - N definition is outside the scope of this course. Readers may convince themselves that the rules are true using intuition.

3.2. Limits of Sequences

Rules for Limits of Sequences

$$(L1) \quad \lim_{n \rightarrow \infty} k = k \quad (\text{where } k \text{ is a constant})$$

$$(L2) \quad \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \quad (\text{where } p \text{ is a positive constant})$$

$$(L3) \quad \lim_{n \rightarrow \infty} \frac{1}{b^n} = 0 \quad (\text{where } b \text{ is a constant greater than } 1)$$

$$(L4) \quad \lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$(L5) \quad \lim_{n \rightarrow \infty} a_n b_n = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$$

$$(L6) \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} \quad \text{provided that } \lim_{n \rightarrow \infty} b_n \neq 0.$$

Remark

- The meaning of (L1) is that if $a_n = k$ for all n where k is a constant, then the sequence $(a_n)_{n=1}^{\infty}$ is convergent and its limit is k .
- The meaning of (L4) is that if both $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are convergent and their limits are L and M respectively, then $(a_n + b_n)_{n=1}^{\infty}$ is also convergent and its limit is $L + M$.
- The following is a special case of (L5). It can be obtained by putting $a_n = k$ for all n and applying (L1).

$$(L5s) \quad \lim_{n \rightarrow \infty} k b_n = k \lim_{n \rightarrow \infty} b_n$$

- Using (L4) and (L5s), we get

$$(L4') \quad \lim_{n \rightarrow \infty} (a_n - b_n) = \lim_{n \rightarrow \infty} a_n - \lim_{n \rightarrow \infty} b_n$$

In fact, Rule (L4) is valid for sum and difference of finitely many sequences. This general result will be referred to as Rule (L4). Similarly, the result for product of finitely many sequences will be referred to as Rule (L5).

In Problem 1 in the last section, the sequence obtained can be represented by the formula $a_n = 4 + \frac{1}{2^n}$. Our intuition tells us that the limit of the sequence is 4. Below we use rules for limits to justify this result.

Example Find $\lim_{n \rightarrow \infty} \left(4 + \frac{1}{2^n}\right)$, if it exists.

Explanation The sequence under consideration is given by $a_n = 4 + \frac{1}{2^n}$. The question asks for the following

- (1) Does the limit of $(a_n)_{n=1}^{\infty}$ exist or not (or equivalently, is the sequence convergent)?
- (2) If the answer to (1) is affirmative, find the limit.

$$\begin{aligned} \text{Solution} \quad \lim_{n \rightarrow \infty} \left(4 + \frac{1}{2^n}\right) &= \lim_{n \rightarrow \infty} 4 + \lim_{n \rightarrow \infty} \frac{1}{2^n} && \text{Rule (L4)} \\ &= 4 + 0 && \text{Rules (L1) and (L3)} \\ &= 4 \end{aligned}$$

□

Remark Below is the logic in the above calculation:

- (1) In the first step, because the constant sequence $(4)_{n=1}^{\infty}$ and the sequence $\left(\frac{1}{2^n}\right)_{n=1}^{\infty}$ are convergent, we can apply Rule (L4).
- (2) The limits of the two sequences are found by Rule (L1) and Rule (L3) respectively in the second step.

The sequence in the next example is the one obtained in Problem 2 in the last section.

Example Find $\lim_{n \rightarrow \infty} \frac{2n^3 - 3n^2 + n}{6n^3}$, if it exists.

Solution

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{2n^3 - 3n^2 + n}{6n^3} &= \lim_{n \rightarrow \infty} \left(\frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2} \right) && \text{Rewrite the expression} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{3} - \lim_{n \rightarrow \infty} \left(\frac{1}{2} \cdot \frac{1}{n} \right) + \lim_{n \rightarrow \infty} \left(\frac{1}{6} \cdot \frac{1}{n^2} \right) && \text{Rule (L4), rewrite 2nd and 3rd terms} \\
 &= \frac{1}{3} - \frac{1}{2} \cdot \lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{6} \cdot \lim_{n \rightarrow \infty} \frac{1}{n^2} && \text{Rules (L1) and (L5s)} \\
 &= \frac{1}{3} - \frac{1}{2} \cdot 0 + \frac{1}{6} \cdot 0 && \text{Rule (L2)} \\
 &= \frac{1}{3}
 \end{aligned}$$

□

Example Find $\lim_{n \rightarrow \infty} (1 + 2n)$, if it exists.

Solution Limit does not exist. This is because we can't find any real number L satisfying the condition that $2n + 1$ is close to L if n is large. □

Remark If we apply rules for limits, we get

$$\begin{aligned}
 \lim_{n \rightarrow \infty} (2n + 1) &= \lim_{n \rightarrow \infty} 2n + \lim_{n \rightarrow \infty} 1 && \text{Rule (L4)} \\
 &= 2 \lim_{n \rightarrow \infty} n + 1 && \text{Rules (L1) and (L5s)}
 \end{aligned}$$

However, we can't proceed because $\lim_{n \rightarrow \infty} n$ does not exist. From this, we see that the given limit does not exist.

FAQ Can we say that $\lim_{n \rightarrow \infty} (1 + 2n)$ is ∞ ?

Answer Limit of a sequence is a *real number* satisfying Condition (*) given in the definition on page 76. Because ∞ is not a real number, we should say that the limit does not exist.

In the next section, we will discuss the meaning of $\lim_{x \rightarrow \infty} f(x) = \infty$ etc. □

Example Find $\lim_{n \rightarrow \infty} \frac{n+1}{2n+1}$, if it exists.

Explanation We can't use Rule (L6) because limits of the numerator and the denominator do not exist. However, we can't conclude from this that the given limit does not exist. To find the limit, we use a trick: *divide the numerator and the denominator by n .*

3.2. Limits of Sequences

$$\begin{aligned}
 \text{Solution } \lim_{n \rightarrow \infty} \frac{n+1}{2n+1} &= \lim_{n \rightarrow \infty} \frac{\frac{n+1}{n}}{\frac{2n+1}{n}} && \text{Divide numerator and denominator by } n \\
 &= \frac{\lim_{n \rightarrow \infty} (1 + \frac{1}{n})}{\lim_{n \rightarrow \infty} (2 + \frac{1}{n})} && \text{Rule (L6), rewrite numerator and denominator} \\
 &= \frac{\lim_{n \rightarrow \infty} 1 + \lim_{n \rightarrow \infty} \frac{1}{n}}{\lim_{n \rightarrow \infty} 2 + \lim_{n \rightarrow \infty} \frac{1}{n}} && \text{Rule (L4)} \\
 &= \frac{1+0}{2+0} && \text{Rules (L1) and (L2)} \\
 &= \frac{1}{2}
 \end{aligned}$$

□

Remark We can apply the following shortcut (called the *Leading Terms Rule*). The method is to throw away the constant term 1 in the numerator and the denominator (*note that if n is very large, compared with n or $2n$, 1 is very small*).

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{n+1}{2n+1} &= \lim_{n \rightarrow \infty} \frac{n}{2n} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{2} = \frac{1}{2}.
 \end{aligned}$$

The Leading Terms Rule for limits of functions at infinity will be discussed in more details in the next section (see page 83).

Exercise 3.2

1. For each of the following, find the limit if it exists.

$$\begin{aligned}
 \text{(a)} \quad \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} & \qquad \text{(b)} \quad \lim_{n \rightarrow \infty} (7 - \frac{5}{n^2}) \\
 \text{(c)} \quad \lim_{n \rightarrow \infty} \frac{3n^2 - 4000}{2n^2 + 10000} & \qquad \text{(d)} \quad \lim_{n \rightarrow \infty} \frac{n^2 - 12345}{n + 1} \\
 \text{(e)} \quad \lim_{n \rightarrow \infty} \frac{5n^2 + 4}{2n^3 + 3} & \qquad \text{(f)} \quad \lim_{n \rightarrow \infty} \left(\frac{1}{n} + (-1)^n \right)
 \end{aligned}$$

2. Suppose \$50,000 is deposited at a bank and the annual interests rate is 2%.

- What amount (correct to the nearest cent) will the account have after one year if interests is
 - compounded quarterly;
 - compounded monthly?
- If interest is compounded n times a year, express the amount A_n after one year in terms of n .
- * Does $\lim_{n \rightarrow \infty} A_n$ exist? What is the value?

- *3. For each of the following sequences $(a_n)_{n=1}^{\infty}$, use computer to find the first 100 (or more) terms. Does $\lim_{n \rightarrow \infty} a_n$ exists? If yes, what is the value?

$$\begin{aligned}
 \text{(a)} \quad a_n &= \left(1 + \frac{1}{n}\right)^n \\
 \text{(b)} \quad a_n &= \left(1 + \frac{2}{n}\right)^n \\
 \text{(c)} \quad a_n &= n \sin \frac{1}{n} \quad (\text{angles are in radians})
 \end{aligned}$$

- *4. Suppose (a_n) is a sequence such that $0 < a_n$ for all n and $a_1 > a_2 > a_3 > \cdots$. Does $\lim_{n \rightarrow \infty} a_n$ exist? What can you tell about the limit?

3.3 Limits of Functions at Infinity

When we consider limit of a sequence $(a_n)_{n=1}^{\infty}$, we let n approach ∞ through the discrete points $n = 1, 2, 3, \dots$. Recall that a sequence means a sequence of real numbers; it is a function from $\mathbb{Z}_+$ to $\mathbb{R}$. In many cases, we will consider functions f from a subset of $\mathbb{R}$ to $\mathbb{R}$ such that $f(x)$ is defined when x is large. For such functions, we can let x approach ∞ continuously (through large real numbers) and consider the behavior of $f(x)$.

Convention A *function* means a function whose domain is a subset of $\mathbb{R}$ and whose codomain is $\mathbb{R}$, unless otherwise stated.

Definition Let f be a function such that $f(x)$ is defined for sufficiently large x . Suppose L is a real number satisfying the following condition:

(*) $f(x)$ is arbitrarily close to L if x is sufficiently large.

Then we say that L is the *limit of f at infinity* and write $\lim_{x \rightarrow \infty} f(x) = L$.

Remark

- The condition “ $f(x)$ is defined for sufficiently large x ” means that there is a real number r such that $f(x)$ is defined for all $x > r$.
- L is called *the* limit because it is unique (if it exists).
- For simplicity, instead of saying Condition (*), we will say

(**) $f(x)$ is close to L if x is large.

Condition (*) means that if we let x increase without bound, then the value $f(x)$ approaches L . To visualize this, imagine a small creature living on the curve $y = f(x)$. Suppose the small creature moves to the right indefinitely. It will get “closer and closer” to the horizontal line $y = L$.

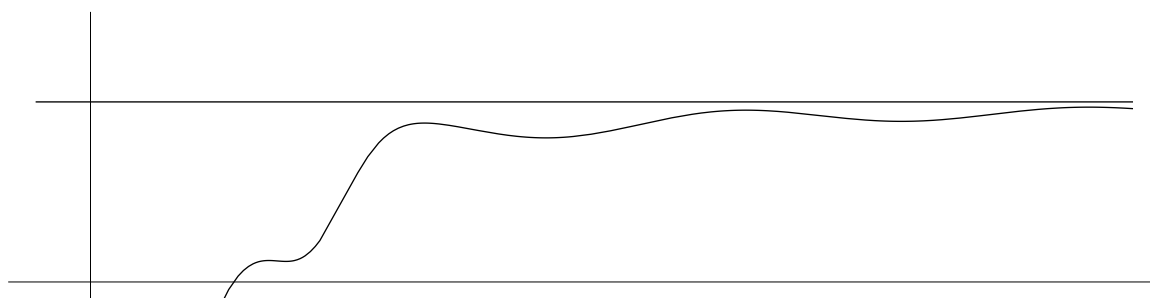


Figure 3.5

FAQ For sequences, we just say “*limit*”. For functions, why are the words “*at infinity*” added?

Answer For functions, there are other types of limits. In Section 3.5, we will discuss limits of functions *at a* (where $a \in \mathbb{R}$). □

3.3. Limits of Functions at Infinity

The following rules for limits of functions at infinity are similar to that for limits of sequences. In (4), (5), (5s) and (6), f and g are functions such that $f(x)$ and $g(x)$ are defined for sufficiently large x .

Rules for Limits of Functions at Infinity

$$(L1) \quad \lim_{x \rightarrow \infty} k = k \quad (\text{where } k \text{ is a constant})$$

$$(L2) \quad \lim_{x \rightarrow \infty} \frac{1}{x^p} = 0 \quad (\text{where } p \text{ is a positive constant})$$

$$(L3) \quad \lim_{x \rightarrow \infty} \frac{1}{b^x} = 0 \quad (\text{where } b \text{ is a constant greater than } 1)$$

$$(L4) \quad \lim_{x \rightarrow \infty} (f(x) \pm g(x)) = \lim_{x \rightarrow \infty} f(x) \pm \lim_{x \rightarrow \infty} g(x)$$

The result is valid for sum and difference of finitely many functions.

$$(L5) \quad \lim_{x \rightarrow \infty} (f(x) \cdot g(x)) = \lim_{x \rightarrow \infty} f(x) \cdot \lim_{x \rightarrow \infty} g(x)$$

The result is valid for product of finitely many functions.

$$(L5s) \quad \lim_{x \rightarrow \infty} (k \cdot g(x)) = k \cdot \lim_{x \rightarrow \infty} g(x)$$

$$(L6) \quad \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow \infty} f(x)}{\lim_{x \rightarrow \infty} g(x)} \quad \text{provided that } \lim_{x \rightarrow \infty} g(x) \neq 0.$$

To consider limits of functions at infinity, we should first check the domains of the functions. For example, if $f(x) = \sqrt{1-x}$, the domain of f is $\{x \in \mathbb{R} : 1-x \geq 0\} = (-\infty, 1]$; it is meaningless to talk about limit of f at infinity. In the next example, the domain of the function $1 - \frac{2}{x^3}$ is $\mathbb{R} \setminus \{0\}$; the function is defined for large x and hence we may consider its limit at infinity (whether the limit exists; and if exists, find the value).

Example Find $\lim_{x \rightarrow \infty} \left(1 - \frac{2}{x^3}\right)$, if it exists.

$$\begin{aligned} \text{Solution} \quad \lim_{x \rightarrow \infty} \left(1 - \frac{2}{x^3}\right) &= \lim_{x \rightarrow \infty} 1 - \lim_{x \rightarrow \infty} \left(2 \cdot \frac{1}{x^3}\right) && \text{Rule (L4), rewrite 2nd term} \\ &= 1 - 2 \cdot \lim_{x \rightarrow \infty} \frac{1}{x^3} && \text{Rules (L1) and (L5s)} \\ &= 1 - 2 \cdot 0 && \text{Rule (L2)} \\ &= 1 \end{aligned}$$

□

Example Find $\lim_{x \rightarrow \infty} (2^{-x} + 3)$, if it exists.

$$\begin{aligned} \text{Solution} \quad \lim_{x \rightarrow \infty} (2^{-x} + 3) &= \lim_{x \rightarrow \infty} 2^{-x} + \lim_{x \rightarrow \infty} 3 && \text{Rule (L4)} \\ &= \lim_{x \rightarrow \infty} \frac{1}{2^x} + 3 && \text{Rewrite first term and Rule (L1)} \\ &= 0 + 3 && \text{Rule (L3)} \\ &= 3 \end{aligned}$$

□

Example Find $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{3x^3 - 4x + 5}$, if it exists.

Explanation Because limits of the numerator and denominator do not exist, we can't apply Rule (6). The first step is to divide the numerator and denominator by x^3 so that the limits at infinity of the new numerator and denominator exist.

Solution

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{x^2 + 1}{3x^3 - 4x + 5} &= \lim_{x \rightarrow \infty} \frac{\frac{x^2 + 1}{x^3}}{\frac{3x^3 - 4x + 5}{x^3}} && \text{Divide numerator and denominator by } x^3 \\
 &= \frac{\lim_{x \rightarrow \infty} \left(\frac{1}{x} + \frac{1}{x^3} \right)}{\lim_{x \rightarrow \infty} \left(3 - \frac{4}{x^2} + \frac{5}{x^3} \right)} && \text{Rule (L6), rewrite numerator and denominator} \\
 &= \frac{\lim_{x \rightarrow \infty} \frac{1}{x} + \lim_{x \rightarrow \infty} \frac{1}{x^3}}{\lim_{x \rightarrow \infty} 3 - \lim_{x \rightarrow \infty} \frac{4}{x^2} + \lim_{x \rightarrow \infty} \frac{5}{x^3}} && \text{Rule (L4)} \\
 &= \frac{0 + 0}{3 - 0 + 0} && \text{Rules (L1), (L2) and (L5s)} \\
 &= 0
 \end{aligned}$$

The next example is similar to the last one. To find limits at infinity for rational functions, we can divide the numerator and denominator by a suitable power of x .

Example Find $\lim_{x \rightarrow \infty} \frac{x^3 + 1}{3x^3 - 4x + 5}$, if it exists.

Solution

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{x^3 + 1}{3x^3 - 4x + 5} &= \lim_{x \rightarrow \infty} \frac{\frac{x^3 + 1}{x^3}}{\frac{3x^3 - 4x + 5}{x^3}} && \text{Divide numerator and denominator by } x^3 \\
 &= \frac{\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^3} \right)}{\lim_{x \rightarrow \infty} \left(3 - \frac{4}{x^2} + \frac{5}{x^3} \right)} && \text{Rule (L6), rewrite numerator and denominator} \\
 &= \frac{\lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{1}{x^3}}{\lim_{x \rightarrow \infty} 3 - \lim_{x \rightarrow \infty} \frac{4}{x^2} + \lim_{x \rightarrow \infty} \frac{5}{x^3}} && \text{Rule (L4)} \\
 &= \frac{1 + 0}{3 - 0 + 0} && \text{Rules (L1), (L2) and (L5s)} \\
 &= \frac{1}{3}
 \end{aligned}$$

To find limits at infinity for rational functions, we can also use the following shortcut.

3.3. Limits of Functions at Infinity

Leading Terms Rule Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ and $g(x) = b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0$, where $a_n \neq 0$ and $b_m \neq 0$. Then we have

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \cdots + b_1 x + b_0} = \lim_{x \rightarrow \infty} \frac{a_n x^n}{b_m x^m}$$

Proof The idea is to extract factor $a_n x^n$ in the numerator and $b_m x^m$ in the denominator. Putting

$$\varphi(x) = \frac{1 + \frac{a_{n-1}}{a_n} \cdot \frac{1}{x} + \frac{a_{n-2}}{a_n} \cdot \frac{1}{x^2} + \cdots + \frac{a_1}{a_n} \cdot \frac{1}{x^{n-1}} + \frac{a_0}{a_n} \cdot \frac{1}{x^n}}{1 + \frac{b_{m-1}}{b_m} \cdot \frac{1}{x} + \frac{b_{m-2}}{b_m} \cdot \frac{1}{x^2} + \cdots + \frac{b_1}{b_m} \cdot \frac{1}{x^{m-1}} + \frac{b_0}{b_m} \cdot \frac{1}{x^m}}$$

we have $\frac{f(x)}{g(x)} = \frac{a_n x^n}{b_m x^m} \cdot \varphi(x)$. It is straightforward to check that $\lim_{x \rightarrow \infty} \varphi(x) = 1$. Hence by Rule (5), we obtain the required result. $\square$

Remark

- (a) If $n = m$, the limit is $\frac{a_n}{b_n}$.
- (b) If $n < m$, the limit is $\lim_{x \rightarrow \infty} \left(\frac{a_n}{b_m} \cdot \frac{1}{x^{m-n}} \right) = 0$.
- (c) If $n > m$, the limit is $\lim_{x \rightarrow \infty} \left(\frac{a_n}{b_m} \cdot x^{n-m} \right)$ which does not exist because as x increases indefinitely, x^{n-m} increases indefinitely.
- The Leading Terms Rule can also be applied to “functions similar to rational functions”, for example, for $f(x) = x + 2\sqrt{x} + 3$ and $g(x) = 5x + 6\sqrt{x} + 7$, we have $\lim_{x \rightarrow \infty} \frac{x + 2\sqrt{x} + 3}{5x + 6\sqrt{x} + 7} = \lim_{x \rightarrow \infty} \frac{x}{5x}$
- The Leading Terms Rule can't be applied to limits of rational functions at a point: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$, where $a \in \mathbb{R}$.

Below we re-do the last two examples using the Leading Terms Rule.

$$\begin{aligned} \textbf{Example} \quad \lim_{x \rightarrow \infty} \frac{x^2 + 1}{3x^3 - 4x + 5} &= \lim_{x \rightarrow \infty} \frac{x^2}{3x^3} && \text{Leading Terms Rule} \\ &= \lim_{x \rightarrow \infty} \left(\frac{1}{3} \cdot \frac{1}{x} \right) && \text{Simplify and rewrite expression} \\ &= \frac{1}{3} \cdot 0 && \text{Rules (L2) and (L5s)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \textbf{Example} \quad \lim_{x \rightarrow \infty} \frac{x^3 + 1}{3x^3 - 4x + 5} &= \lim_{x \rightarrow \infty} \frac{x^3}{3x^3} && \text{Leading Terms Rule} \\ &= \lim_{x \rightarrow \infty} \frac{1}{3} && \text{Simplify expression} \\ &= \frac{1}{3} && \text{Rule (L1)} \end{aligned}$$

In the next example, the function can be considered as a product or a quotient of two functions. However, we can't apply Rule (5) or (6) because limit at infinity of one of the functions does not exist. To find the limit, we need the following result.

Sandwich Theorem Let f , g and h be functions such that $f(x)$, $g(x)$ and $h(x)$ are defined for sufficiently large x . Suppose that $f(x) \leq g(x) \leq h(x)$ if x is sufficiently large and that both $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow \infty} h(x)$ exist and are equal (with common limit denoted by L). Then we have $\lim_{x \rightarrow \infty} g(x) = L$.

Remark The condition “ $f(x) \leq g(x) \leq h(x)$ if x is sufficiently large” means that there is a real number r such that the inequalities are true for all $x > r$.

Example Find $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$, if it exists.

Explanation The given function can be written as a product of two functions: $\sin x$ and $\frac{1}{x}$. For the second function, its limit at infinity is 0. However, for the first function, its limit at infinity does not exist. Thus we can't apply Rule (5).

Solution Since $-1 \leq \sin x \leq 1$ for all real numbers x , it follows that

$$\frac{-1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x} \quad \text{for all } x > 0.$$

Note that $\lim_{x \rightarrow \infty} \frac{-1}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Thus by the Sandwich Theorem, we have $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$. □

Example Find $\lim_{x \rightarrow \infty} (1 + \log x)$, if it exists.

Remark Since $\lim_{x \rightarrow \infty} \log x$ does not exist, we can't apply Rule (L4).

Solution $\lim_{x \rightarrow \infty} (1 + \log x)$ does not exist. This is because if x increases without bound, so does $1 + \log x$.

Infinite Limits

In the last example, although limit does not exist, we know that if x increases indefinitely, so does $(1 + \log x)$. In the limit notation $\lim_{x \rightarrow \infty}$, the symbol $x \rightarrow \infty$ indicates that “ x increases indefinitely”, or “ x approaches ∞ ”. Using the same idea, we also write $1 + \log x \rightarrow \infty$ which indicates that the value increases indefinitely (as x increases indefinitely). Putting $y = 1 + \log x$, we write $y \rightarrow \infty$ as $x \rightarrow \infty$. Concerning the graph of $y = f(x)$ in the coordinate plane, $x \rightarrow \infty$ means that x goes to the right indefinitely, approaching the point ∞ (an imaginary point on the right) and $y \rightarrow \infty$ means that y goes up indefinitely, approaching the point ∞ (an imaginary point at the top).

Notation Let f be a function such that $f(x)$ is defined for sufficiently large x . Suppose that

(*) $f(x)$ is arbitrarily large if x is sufficiently large.

Then we write $\lim_{x \rightarrow \infty} f(x) = \infty$.

Remark

- Because ∞ is not a real number, $\lim_{x \rightarrow \infty} f(x) = \infty$ does not mean the limit exists. In fact, it indicates that the limit does not exist and explains why it does not exist.

3.3. Limits of Functions at Infinity

- Instead of $\lim_{x \rightarrow \infty} f(x) = \infty$, we also write $f(x) \rightarrow \infty$ as $x \rightarrow \infty$.
- For simplicity, instead of saying Condition (*), we will say
(**) $f(x)$ is large if x is large.
- Similar to $\lim_{x \rightarrow \infty} f(x) = \infty$, we also have $\lim_{x \rightarrow \infty} f(x) = -\infty$ which means that
(**) $f(x)$ is large negative if x is large.

Example (a) $\lim_{x \rightarrow \infty} (1 + x^2) = \infty$ (limit does not exist) (b) $\lim_{x \rightarrow \infty} (1 - x^2) = -\infty$ (limit does not exist)

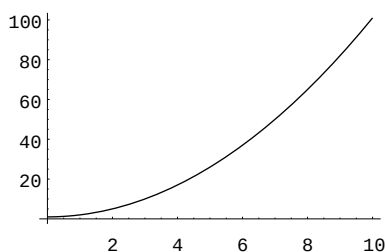


Figure 3.6

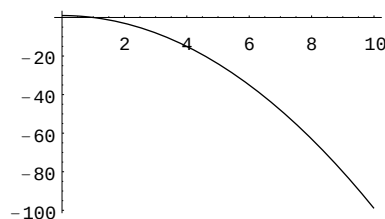


Figure 3.7

Example $\lim_{x \rightarrow \infty} \frac{1 + x^2}{1 + x} = \lim_{x \rightarrow \infty} \frac{x^2}{x}$ Leading Terms Rule
 $= \lim_{x \rightarrow \infty} x$
 $= \infty$ (limit does not exist)

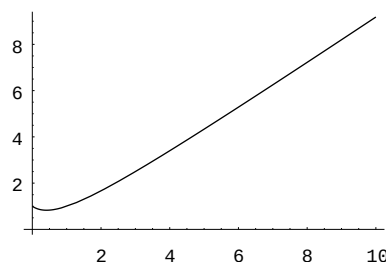


Figure 3.8

Limits at negative infinity

Similar to limits at infinity, we may consider *limits at negative infinity* provided that $f(x)$ is defined for x sufficiently large negative. Readers can figure out the meaning of the following notations:

- $\lim_{x \rightarrow -\infty} f(x) = L$ where L is a real number;
- $\lim_{x \rightarrow -\infty} f(x) = \infty$;
- $\lim_{x \rightarrow -\infty} f(x) = -\infty$.

Example

- (1) $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$
- (2) $\lim_{x \rightarrow -\infty} x^3 = -\infty$ (limit does not exist)
- (3) $\lim_{x \rightarrow -\infty} (1 - x^3) = \infty$ (limit does not exist)

FAQ Can we perform addition, multiplication etc. with ∞ or $-\infty$?

Answer Yes and no. For example,

- $1 + \infty = \infty$,

Exercise 3.3

1. For each of the following, find the limit if it exists.

(a) $\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x-1}}$

(b) $\lim_{x \rightarrow \infty} (15 - 16x^{-3})$

(c) $\lim_{x \rightarrow \infty} 5^{-x}$

(d) $\lim_{x \rightarrow \infty} \sqrt{x}$

(e) $\lim_{x \rightarrow \infty} \frac{x^2 + 9}{x^3 + 1}$

(f) $\lim_{x \rightarrow \infty} \frac{x^2 + 9}{x^2 + 1}$

(g) $\lim_{x \rightarrow \infty} \frac{x^2 + 9}{x + 1}$

(h) $\lim_{x \rightarrow \infty} \frac{|x|}{x}$

(i) $\lim_{x \rightarrow -\infty} \frac{|x|}{x}$

(j) $\lim_{x \rightarrow -\infty} x \sin x$

2. The concentration C of a drug in a patient's bloodstream t hours after it was injected is given by

$$C(t) = \frac{0.15t}{t^2 + 3}.$$

(a) Find $\lim_{t \rightarrow \infty} C(t)$.

(b) Interpret the result in (a).

3. The population P of a certain small town t years from now is predicted to be $P(t) = 35000 + \frac{10000}{(t+2)^2}$.

(a) Find the population in the long run.

(b) Use computer to sketch the graph of P . What can you tell from the graph?

*4. For each of the following functions f , use computer to find $f(x)$ for large x . Guess whether $\lim_{x \rightarrow \infty} f(x)$ exists or not. If the limit exists, what is the limit?

(a) $f(x) = \sqrt{x+1} - \sqrt{x}$

(b) $f(x) = \sqrt{x^2 + x} - x$

(c) $f(x) = \frac{x^{99}}{2^x}$

Rules for Limits of Functions at a Point

$$(La1) \quad \lim_{x \rightarrow a} k = k \quad (\text{where } a \in \mathbb{R} \text{ and } k \text{ is a constant})$$

$$(La2) \quad \lim_{x \rightarrow a} x^n = a^n \quad (\text{where } a \in \mathbb{R} \text{ and } n \text{ is a positive integer})$$

$$(La2') \quad \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a} \quad (\text{where } a \in \mathbb{R} \text{ and } n \text{ is an odd positive integer})$$

$$\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a} \quad (\text{where } 0 < a \in \mathbb{R} \text{ and } n \text{ is an even positive integer})$$

$$(La3) \quad \lim_{x \rightarrow a} b^x = b^a \quad (\text{where } a \in \mathbb{R} \text{ and } b \text{ is a positive real number})$$

$$(La4) \quad \lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

The result is valid for sum and difference of finitely many functions.

$$(La5) \quad \lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

The result is valid for product of finitely many functions.

$$(La5s) \quad \lim_{x \rightarrow a} (k \cdot g(x)) = k \cdot \lim_{x \rightarrow a} g(x) \quad (\text{where } k \text{ is a constant})$$

$$(La6) \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{provided that } \lim_{x \rightarrow a} g(x) \neq 0.$$

Example Find $\lim_{x \rightarrow 4} (1 + x^2)$, if it exists.

$$\begin{aligned} \text{Solution} \quad \lim_{x \rightarrow 4} (1 + x^2) &= \lim_{x \rightarrow 4} 1 + \lim_{x \rightarrow 4} x^2 && \text{Rule (La4)} \\ &= 1 + 4^2 && \text{Rules (La1) and (La2)} \\ &= 17 \end{aligned}$$

Recall that a polynomial function p is a function that can be written in the following form

$$p(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0,$$

3.5. Two-sided Limits

where $c_0, c_1, \dots, c_n$ are constants. Using the method in the above example, we can prove the following theorem which means that the limit of a polynomial function at any real number can be found by substitution.

Theorem 3.5.1 *Let $p(x)$ be a polynomial and let a be a real number. Then we have*

$$\lim_{x \rightarrow a} p(x) = p(a).$$

Example Find $\lim_{x \rightarrow 2} \frac{x-1}{x^2+x-2}$, if it exists.

$$\begin{aligned} \text{Solution } \lim_{x \rightarrow 2} \frac{x-1}{x^2+x-2} &= \frac{\lim_{x \rightarrow 2} (x-1)}{\lim_{x \rightarrow 2} (x^2+x-2)} && \text{Rule (La6)} \\ &= \frac{2-1}{2^2+2-2} && \text{Theorem 3.5.1} \\ &= \frac{1}{4} \end{aligned}$$

Recall that a rational function r is a function that can be written in the form

$$r(x) = \frac{p(x)}{q(x)},$$

where p and q are polynomial functions. Using the method in the above example, we can prove the following theorem which means that the limit of a rational function at any a belonging to its domain can be found by substitution.

Theorem 3.5.2 *Let $p(x)$ and $q(x)$ be polynomials and let a be a real number. Suppose that $q(a) \neq 0$. Then we have*

$$\lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}.$$

Example Find $\lim_{x \rightarrow 1} \frac{x-1}{x^2+x-2}$, if it exists.

Explanation The rational function $f(x) = \frac{x-1}{x^2+x-2}$ is undefined at $x = 1$. This means that 1 does not belong to the domain of f and so we can't apply Theorem 3.5.2. If we substitute $x = 1$ into the numerator and denominator, we get $\frac{0}{0}$. We say that the limit is in the *indeterminate form* $\frac{0}{0}$.

To find the limit, we replace f by a function g which coincides with f on the left-side and right-side of 1 such that the limit of g at 1 can be found by substitution (see the following figures).

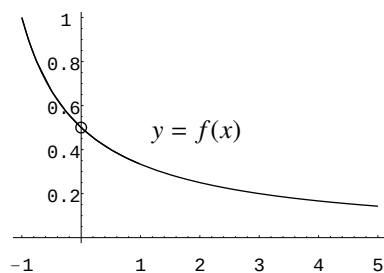


Figure 3.18(a)

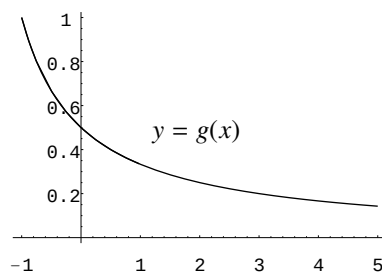


Figure 3.18(b)

The function g is given by $g(x) = \frac{1}{x+2}$. It can be found by simplifying the expression defining f :

$$f(x) = \frac{x-1}{(x-1)(x+2)} = g(x) \quad \text{for all } x \in \mathbb{R} \setminus \{1, -2\}.$$

$$\begin{aligned} \text{Solution} \quad \lim_{x \rightarrow 1} \frac{x-1}{x^2+x-2} &= \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+2)} \\ &= \lim_{x \rightarrow 1} \frac{1}{x+2} && \text{Simplify expression} \\ &= \frac{1}{1+2} && \text{Theorem 3.5.2} \\ &= \frac{1}{3} \end{aligned}$$

Example Find $\lim_{x \rightarrow 1} \frac{x+1}{x^2+x-2}$, if it exists.

Explanation The rational function $f(x) = \frac{x+1}{x^2+x-2}$ is undefined at $x = 1$. Thus we can't use Theorem 3.5.2. If we put $x = 1$ into the numerator and denominator, we get $\frac{2}{0}$. Limits (left-side and right-side) of the form $\frac{\text{non-zero number}}{0}$ are ∞ or $-\infty$. See the solution below for more details.

Solution $\lim_{x \rightarrow 1} \frac{x+1}{x^2+x-2}$ does not exist. This is because if x is close to 1, the numerator is close to 2 whereas the denominator is close to 0 and so the fraction is very large in magnitude (may be positive or negative). $\square$

Remark Indeed, the denominator is $x^2 + x - 2 = (x-1)(x+2)$. Therefore, if x is close to and greater than 1, the denominator is small positive. Hence we have $\lim_{x \rightarrow 1+} \frac{x+1}{x^2+x-2} = \infty$. Similarly we have $\lim_{x \rightarrow 1-} \frac{x+1}{x^2+x-2} = -\infty$.

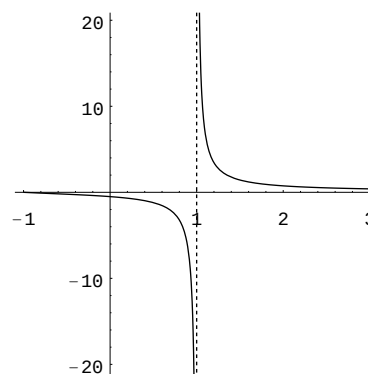


Figure 3.19

Example Let $f(x) = x^2 + 3$. Find $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

Explanation The expression $\frac{f(x+h) - f(x)}{h}$ involves two variables x and h . However, the question asks for $\lim_{h \rightarrow 0}$ (limit of the expression as h approaches 0). This implies that x is considered as a constant. In this way, the expression $\frac{f(x+h) - f(x)}{h}$ is considered as a function of h , defined for all $h \neq 0$. The limit is in the indeterminate form $\frac{0}{0}$ because if we put $h = 0$ in the expression, the numerator and denominator are both 0. To find the limit, we simplify the expression so that the troublesome factor h in the denominator is canceled.

$$\begin{aligned} \text{Solution} \quad \frac{f(x+h) - f(x)}{h} &= \frac{((x+h)^2 + 3) - (x^2 + 3)}{h} \\ &= \frac{(x^2 + 2xh + h^2 + 3) - (x^2 + 3)}{h} \\ &= \frac{2xh + h^2}{h} \\ &= \frac{h(2x + h)}{h} \\ &= 2x + h \end{aligned}$$

3.5. Two-sided Limits

Since $\frac{f(x+h) - f(x)}{h}$ and $2x + h$ (considered as functions of h) are equal on the left-side and right-side of $h = 0$, it follows that

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0} (2x + h) \\ &= 2x + 0 && \text{Theorem 3.5.1} \\ &= 2x\end{aligned}$$

Remark The expression $\frac{f(x+h) - f(x)}{h}$ is called a *difference quotient*. Limits of difference quotients will be discussed in detail in Chapter 4.

Summary for Limits In this chapter, we have introduced the following types of limits:

$$\lim_{n \rightarrow \infty} a_n, \quad \lim_{x \rightarrow \infty} f(x), \quad \lim_{x \rightarrow -\infty} f(x), \quad \lim_{x \rightarrow a-} f(x), \quad \lim_{x \rightarrow a+} f(x), \quad \lim_{x \rightarrow a} f(x).$$

Since the definitions for $\lim_{n \rightarrow \infty} a_n$ and $\lim_{x \rightarrow \infty} f(x)$ are similar, we will omit limits of sequences in the following discussion. Note that for functions, the five types of limits take the form

$$\lim_{x \rightarrow \square} f(x)$$

where $\square$ can be ∞ , $-\infty$, $a-$, $a+$ or a . The notation

$$\lim_{x \rightarrow \square} f(x) = L,$$

where L is a real number, means the following

$f(x)$ is arbitrarily close to L if x is “sufficiently close to (and different from)” $\square$,

which, in short, is written as

$f(x)$ is close to L if x is “close to (and different from)” $\square$,

where $f(x)$ is close to L has the usual meaning and

- x is “close to (and different from)” ∞ means x is large;
- x is “close to (and different from)” $-\infty$ means x is large negative;
- x is “close to (and different from)” a has the usual meaning.

If we cannot find a real number L such that $\lim_{x \rightarrow \square} f(x) = L$, then we say that $\lim_{x \rightarrow \square} f(x)$ does not exist. There are several possibilities for this. We may have

$$\lim_{x \rightarrow \square} f(x) = \infty \quad \text{or} \quad \lim_{x \rightarrow \square} f(x) = -\infty \quad \text{or} \quad \text{other behavior such as oscillation.}$$

Similar to the above discussion, the notation $\lim_{x \rightarrow \square} f(x) = \infty$ means that $f(x)$ is “close to” ∞ if x is “close to (and different from)” $\square$, where “close to” ∞ means large.

Exercise 3.5

1. For each of the following, find the limit, if it exists.

(a) $\lim_{x \rightarrow 2} (x^2 + 3x - 4)$

(b) $\lim_{x \rightarrow 7} (x^2 - 5x - 8)^3$

(c) $\lim_{x \rightarrow 0} \left(\frac{3x-5}{2x+7} \right)^2$

(d) $\lim_{x \rightarrow 1} \frac{x\sqrt{x^2+1}}{x+1}$

(e) $\lim_{x \rightarrow -2} 2^{5+4x}$

(f) $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x^2 + x - 2}$

(g) $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x^2 - x + 2}$

(h) $\lim_{x \rightarrow 6} \frac{x^2 - 6x}{x^2 - 5x - 6}$

(i) $\lim_{x \rightarrow 5} \frac{x^2 + 25}{x - 5}$

(j) $\lim_{x \rightarrow 3} \frac{x - 3}{x^2 + 9}$

(k) $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$

(l) $\lim_{x \rightarrow 2} \frac{x - 5}{2 - \sqrt{x - 1}}$

2. For each of the following f , find $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

(a) $f(x) = 4x - 13$

(b) $f(x) = x^3$

(c) $f(x) = \frac{1}{x}$

(d) $f(x) = \sqrt{x}$